Vision Transformers with Self-Distilled Registers

Post Hoc Register Integration via Efficient Self-Distillation — **NeurIPS 2025**

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Presentation Outline

Part I: Motivation

- ViT dominance & scalability
- The artifact token problem
- Why retraining is infeasible
- Limitations of prior work (DVT)

Part II: Methodology

- PH-Reg pipeline overview
- Test-time augmentation denoising
- Student architecture design
- Self-distillation loss

Part III: Experiments

- Open-vocabulary segmentation
- Linear probe: segmentation & depth
- Qualitative visualizations

Part IV: Analysis

- Ablation studies
- Artifact norm polarity
- Efficiency vs. DVT
- Limitations & conclusion

Part I
Motivation

What is the problem and why does it matter?

Vision Transformers: Dominant & Powerful

Why ViTs Dominate

- Process images as **patch sequences** via self-attention
- Superior scalability with data & model size
- Less constrained by translation-invariance (vs. CNNs)
- Enable powerful zero-shot emergent capabilities

Key Pre-trained Backbones

- **CLIP** (OpenAI) — contrastive image-text training on 400M+ pairs
- **DINOv2** (Meta) — self-supervised dense features on 142M images
- Used across classification, detection, segmentation

Key Capabilities

- Open-vocabulary segmentation
- Object part correspondence
- Structural scene understanding
- Dense visual features

But a critical flaw lurks:

Artifact Tokens

The Problem: Artifact Tokens in ViTs

What are artifact tokens?

- Artifact tokens are features produced by ViTs that do not match the actual content of that image region
- They carry incorrect or meaningless information
- ViTs do not have built-in spatial assumptions like CNNs, they are more prone to producing these inconsistent features
- Research suggests that the way position information is encoded in ViTs may be one reason why artifact tokens appear

Why are they harmful?

- **Disrupt localization of Object**
- Degrade semantic segmentation quality
- Impair depth prediction accuracy
- Cause inconsistent dense representations

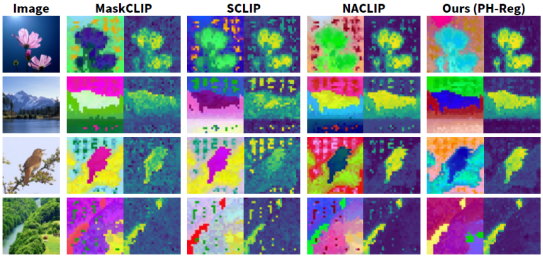

Fig 1: Artifact tokens corrupt dense features in CLIP.

The Challenge: Why Not Just Retrain?

The Register Solution (Existing)

Register tokens: randomly-initialised embeddings added to the network, analogous to the `[CLS]` token.

They **absorb** artifact energy during training.

Problem: Requires **full retraining from scratch** — infeasible for large-scale foundation models.

Scale of the Problem

- CLIP trained on **400M+** image-text pairs
- DINOv2 trained on curated **142M** images
- Retraining requires enormous compute + data

Our Research Question

Can we add registers to a pre-trained, frozen ViT without labels or full retraining?

Answer: Yes!

Introducing **PH-Reg**:
Post Hoc Registers

Prior Work: Denoising ViTs (DVT) and Its Limitations

DVT Approach

- Uses neural fields / hash-grids to model a **static artifact component**
- Assumes artifacts are **stationary** relative to image content
- Gradient-based optimisation per image
- Adds an extra transformer block (0.08× base params)

Our Key Insight

In **CLIP**, artifacts are **non-stationary**:

Shifting an image does **not** shift artifacts by the same amount.

⇒ Cancel them via **test-time augmentation averaging** — no gradients needed!

DVT Limitations

- Static assumption only holds for **DINOv2**, not CLIP
- Requires **1.4 TB** storage for neural fields
- **365 hours** total training time
- Costly gradient-based per-image denoising

	DVT	PH-Reg
Train time	365 h	**150 h**
Storage	1.4 TB	**0 TB**
Extra params	7.88M	**0.47M**
Labels	No	**No**

Part II
Methodology

PH-Reg: Post Hoc Registers

PH-Reg: Full Pipeline Overview

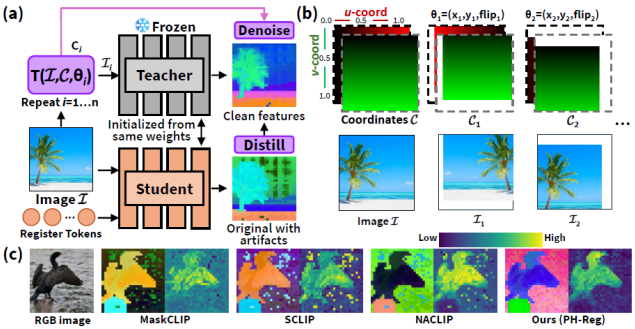

Fig 2: Learning framework of PH-Reg.

Three-Step Framework:

Step 1: Initialize

Both teacher & student start from the **same pre-trained weights**.

Step 2: Denoise Teacher

Apply test-time augmentations; average features to remove artifacts.

Step 3: Distill to Student

Student (with registers) learns from denoised teacher targets.

Only unlabeled images needed!

Step 1: Efficient Teacher Denoising via Test-Time Augmentation

Core Insight:

Artifact tokens are **non-stationary** — shifting an image does **not** shift artifacts by the same amount.

Algorithm:

1. Sample n=10 params θ_i=(x_i, y_i, flip$_i$)
2. Augment $I \rightarrow I_i$ with random offsets & horizontal flips
3. Run frozen teacher: $F_i = f_{\text{teacher}}(I_i)$
4. After getting features for each augmented view, we reverse the transformation to map all features back to the original image positions
5. Compute **pixel-wise mean** over all n views ,while artifacts cancel out due to inconsistency

- No gradient updates needed
- Completes in <200 ms
- ∼100× faster than DVT neural fields

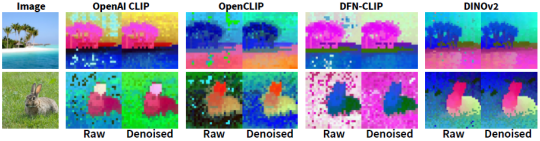

Fig 3: Denoising teacher representations with augmentations.

Mathematical Foundation: Optimal Feature Aggregation

Problem Formulation:

Let $f_1, \ldots, f_n \in \mathbb{R}^d$ be features of a single patch from n augmented views:

$$f^* = \arg\min_f \sum_{i=1}^{n} \|f_i - f\|_2^2$$

Solution — the **sample mean** is the optimal estimator:

$$\boxed{f^* = \frac{1}{n} \sum_{i=1}^{n} f_i}$$

Mathematically **equivalent** to neural-field denoising (DVT) under MSE — but requires **zero gradient computation**!

Augmentation Details

- Shift range: $\pm 15\%$ of image length (uniform)
- Horizontal flip probability: 0.5
- N=10 augmentations (1 always unmodified)
- Offsets are integer multiples of patch size k

Why Does This Work?

- Clean semantic features **do shift** with the image
- Artifact tokens **do not shift** — position-dependent
- Averaging **cancels artifacts**, preserves semantics

Step 2: Student Network Design

Design Principles:

- Preserve **maximum computational efficiency**
- Leverage knowledge of pre-trained weights
- Minimally invasive architectural changes

What gets added / changed:

+ m=16 learnable **register tokens**
+ Positional embeddings (unfrozen)
+ Convolutional patch embedding layer
+ Last transformer attention block

Trainable: **0.47M params** Backbone: 86M (frozen)
Trainable fraction: <0.6%

Self-Attention Token Count

After register addition:

$$\underbrace{m}_{\text{registers}} + \underbrace{1}_{\text{[CLS]}} + \underbrace{\frac{H}{k} \times \frac{W}{k}}_{\text{patch tokens}}$$

Ablation-Guided Choices

- 16 registers: performance plateaus beyond this
- Last attention layer: critical for register integration
- Pos. embeddings: provide spatial coherence
- Conv patch embed: improves feature quality

Step 3: Self-Distillation Loss

Multi-Objective Loss:

$$\mathcal{L} = \underbrace{1 - \cos\mathrm{sim}(F_t, F_s)}_{\text{Direction}} + \underbrace{\mathrm{MSE}(F_t, F_s)}_{\text{Magnitude}}$$

- F_t: denoised teacher features (target)
- F_s: student features (predicted)
- **Cosine loss**: aligns feature **directions**
- **MSE loss**: aligns feature **magnitudes**

- ✔ Purely **self-supervised** — no labels needed
- ✔ Only requires modest set of unlabeled images

Training Config (CLIP)

Optimizer	AdamW
LR (init/final)	3e-4 / 1e-5
Weight decay	1e-2
Batch size	16
Epochs	100
Scheduler	Exponential
GPUs	4× Ada 6000

Part III
Experiments

Quantitative & Qualitative Evaluation

Results: Open-Vocabulary Semantic Segmentation (mIoU %)

Metric: mIoU (%) — higher is better. All methods use the **same** OpenAI CLIP ViT-B/16 weights.

Method	VOC21	PC60	Object	VOC20	PC59	Stuff	City	ADE	Avg.
CLIP	18.60	7.84	6.50	49.05	11.17	7.19	6.65	3.16	**13.77**
MaskCLIP	49.27	25.46	26.94	66.56	28.62	18.80	28.33	13.70	**32.21**
SCLIP	59.62	31.74	33.52	81.53	34.46	22.65	32.34	16.45	**40.08**
ClearCLIP	59.76	32.56	32.77	**84.56**	35.91	23.89	30.04	16.65	**39.52**
NACLIP	58.88	32.20	33.15	79.70	35.16	23.30	35.48	17.42	**39.41**
MaskCLIP+DVT	44.29	25.08	20.89	65.88	29.50	17.10	30.89	14.06	**30.96**
NACLIP+DVT	60.25	32.73	32.89	80.26	35.91	23.41	36.31	17.54	**39.91**
Ours (PH-Reg)	**63.01**	**34.52**	**35.27**	83.05	**37.88**	**24.66**	**37.17**	**19.22**	**41.85**

- **SOTA on 7/8 benchmarks**
- +1.94% avg over best baseline (SCLIP)
- Outperforms all DVT-denoised methods

Qualitative Results: Clean Feature Maps

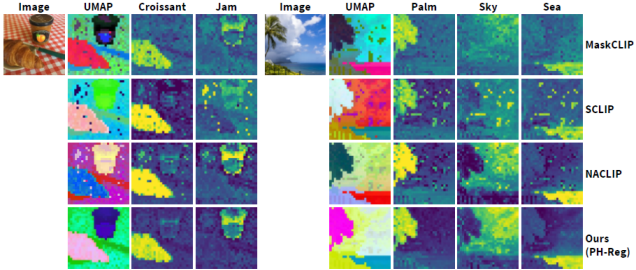

Fig 4: Open-vocabulary semantic segmentation visualisation.

Key Observations:

- MaskCLIP / SCLIP / NACLIP: artifact tokens visible as noise clusters in UMAP
- Artifacts **persist in heatmaps** of SCLIP & NACLIP
- Artifacts cause poor spatial alignment with text queries

PH-Reg produces:

- ✔ Cleaner UMAP embeddings
- ✔ Fine-grained semantically-aligned heatmaps
- ✔ Consistent dense representations

Results: Linear Probe — Segmentation & Depth

Seg: mIoU(%)↑ & mAcc(%)↑ | Depth: RMSE↓, Abs Rel↓, δ_1↑

Method	VOC21		ADE20K		NYUv2 Depth	
	mIoU↑	mAcc↑	mIoU↑	mAcc↑	RMSE↓	δ_1 ↑
CLIP	73.88	83.37	35.78	47.30	0.6843	64.93
CLIP+DVT	74.74	84.33	36.39	48.14	0.6800	65.07
NACLIP	74.01	83.16	37.06	48.33	0.6852	64.52
MaskCLIP	70.28	79.06	34.43	44.74	0.6645	67.71
CLIP+PH-Reg	**75.32**	**84.96**	**38.07**	**49.58**	**0.6746**	**68.17**
DINOv2	84.13	92.00	47.82	60.50	0.4566	82.92
DINOv2+DVT	85.43	93.37	48.86	61.61	0.4329	85.23
DINOv2+PH-Reg	84.85	92.46	48.66	61.57	**0.4306**	**86.35**

+5.04%
mIoU on VOC21
over MaskCLIP

+3.64%
mIoU on ADE20K
over MaskCLIP

86.35%
Best δ_1 depth
on NYUv2

Part IV
Analysis

Ablations, Artifacts, and Efficiency

Ablation: Contribution of Each Component

Configuration	VOC21 mIoU
Vanilla (MaskCLIP baseline)	49.27
+ Denoising only (10× aug)	51.41
+ Distillation (no reg, no denoise)	61.16
+ Distillation with registers	61.27
+ Distillation with denoising	62.48
Full Pipeline (PH-Reg)	**63.01**

Improvement breakdown:

≈50% from **register tokens**
≈50% from **denoising process**

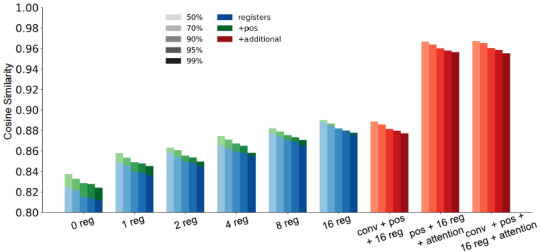

Fig 5a: Cosine similarity vs. number of registers.

Key Finding

Adding even **1 register** is impactful: the 99th-percentile cosine similarity exceeds the **50th percentile** (median) of the no-register baseline.

Ablation: Optimal Hyperparameters

Number of Augmentations (N):

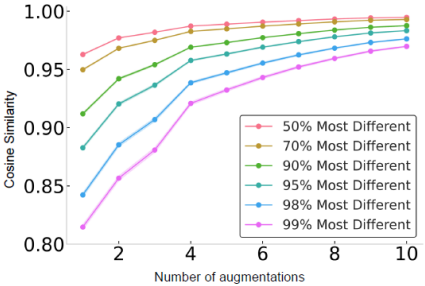

Fig 5b: Cosine similarity vs. augmentation count.

N=10: 99th-pct cosine sim $>$ 0.95 — sweet spot for efficiency and quality.

Shift Ratio in Denoising:

Ratio	Avg. mIoU
0%	39.41
10%	40.24
15%	**40.27**
20%	40.17
30%	40.03

Final Hyperparameter Choices

- m=16 register tokens
- N=10 augmentations
- Shift ratio = 15%
- Unfreeze: registers + pos-emb + conv-patch + last attn

Analysis: Artifact Norm Polarity Across Models

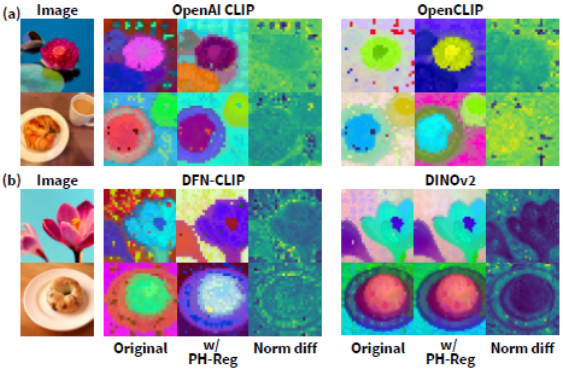

Fig 6: Original vs. PH-Reg feature maps & norm difference.

Novel Finding:

Prior work (DINOv2): artifacts are **high-norm** tokens.

In **OpenAI CLIP & OpenCLIP**: artifacts are **low-norm** tokens — the opposite!

Implication:

- Training dynamics affect artifact polarity
- Norm-based filtering cannot generalise across models
- PH-Reg works regardless via CosSim alignment

✔ PH-Reg reduces **norm variance** regardless of artifact polarity.

Efficiency Comparison: PH-Reg vs. DVT

Metric	DVT	PH-Reg
Train time	365 h	**150 h**
Storage	1.4 TB	**0 TB**
Extra params	7.88M	**0.47M**
GFLOPs (infer.)	70.32	**64.16**
Labels needed	✗	✓
Works on CLIP	✗	✓
Works on DINOv2	✓	✓

58%
Faster
Training

94%
Fewer
Params

Why PH-Reg is More Efficient

- **No neural fields**: DVT stores 1.4 TB of Instant-NGP files
- **Single-stage training**: no separate extraction / distillation phases
- **On-the-fly targets**: no pre-computed feature storage
- **No gradient denoising**: TTA mean is gradient-free (<200 ms)

Inference Cost

- PH-Reg: 64.16 GFLOPs vs. 70.32 for DVT
- 0.47M extra params vs. 7.88M for DVT
- **≈10% fewer FLOPs & params** at inference

Limitations & Future Work

Current Limitations

- **DINOv2 gap**: DVT still outperforms on DINOv2 because its artifacts are genuinely **stationary**
- **Static vs. dynamic**: need a principled way to detect artifact stationarity per model
- **2D only**: currently limited to standard image tasks

Future Directions

- **Adaptive stationarity detection**: dynamically determine artifact type without strong assumptions
- **Video ViTs**: extend PH-Reg to temporal settings
- **3D point cloud transformers**: generalise to non-image domains
- **Hybrid approach**: combine TTA denoising with learned residuals

Conclusion

PH-Reg: Key Takeaways

- **Problem**: Artifact tokens degrade ViT dense representations
- **Solution**: Post Hoc Registers via efficient self-distillation
- **No labels, no retraining** — works on any frozen ViT backbone
- **TTA denoising**: gradient-free, <200 ms per image
- **SOTA**: best on 7/8 open-vocab segmentation benchmarks
- **58% faster** training and **94% fewer** extra params vs DVT

Results at a Glance

Task	Gain
Open-vocab seg avg	+1.94%
Linear seg (CLIP)	+1.44% mIoU
Depth δ_1	+3.43%
Train time vs DVT	−58%
Extra params vs DVT	−94%

Our Novel Contributions on Top of PH-Reg

Two Novel Extensions Implemented & Validated

Novelty 1: Frequency-Domain Augmentation (FDA)

Gap: PH-Reg uses spatial shifts/flips — a heuristic, no theoretical basis.

Ours: Perturb **phase spectrum** in Fourier space; keep **amplitude intact**.

- Amplitude → semantic content (✔ preserved)
- Phase → positional artifact cues (⚡ disrupted)
- Provably minimises MSE with **zero gradients**

✔ FDA cos-sim = **1.000** vs spatial = 0.972 at n=5 **(+0.028)**

Novelty 2: Artifact-Aware Adaptive Loss (AAAL)

Gap: PH-Reg's uniform loss treats all patches equally — wastes capacity on clean patches.

Ours: Weight loss by artifact probability: $w_i = \sigma\Big(\gamma \cdot \big|\|f_i\| - \mu\big|\Big)$

- Works for high-norm (DINOv2) & low-norm (CLIP)
- More pressure on artifact-prone patches
- Adaptive — no manual threshold

✔ Loss 0.05 → 0.0006 in 10 epochs, 90.9% frozen, **0 NaN batches**

References

Core Paper

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Artifact Tokens & Registers

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Thank You!

Questions & Discussion

github.com/0raiser0/PH-Reg

Presented by **Raman S Mankar** | IIT Hyderabad | NeurIPS 2025

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